Reproducing "Mapping Heat in the U.S. Financial System" from Public Data

A self-contained replication guide

October 2026

Abstract

The Federal Reserve FEDS Notes *Mapping Heat in the U.S. Financial System* (August 2015) builds a heat map of financial-system vulnerabilities from 44 indicators grouped into 14 components. This paper documents a full replication of that exercise from public data alone. Every indicator is rebuilt from a named public source with its exact construction given; the index pipeline is specified formula by formula; and the results are scored against the paper's own published "accessible materials" tables. 43 of 44 indicators are replicated with public proxies (transformed-signal correlations 0.37–1.00, median 0.91); the documented pipeline reproduces the published component indices at correlations 0.85–0.98 for 13 of 14 components (aggregate 0.97), and the end-to-end public-proxy panel tracks the published aggregate index at 0.93. Two orientation corrections to the paper's data appendix were required, both discovered empirically. The one indicator with no workable public proxy (the weighted-average maturity gap at commercial banks) and the one component the documented pipeline cannot reproduce even on the paper's own data (commercial real estate) are documented as findings. A reader with only this document and internet access can reproduce the entire exercise.

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1 The original paper

The Federal Reserve FEDS Notes *Mapping Heat in the U.S. Financial System* (August 2015; data through 2015Q1, sample 1990Q1–2014Q4) proposes a framework for assessing the build-up of vulnerabilities in the U.S. financial system. It collects **44 indicators** of financial and balance-sheet conditions, groups them into **14 components** spanning three broad vulnerability categories—valuation pressures, nonfinancial-sector borrowing, and financial-sector health—and develops an algorithmic monitoring approach meant to complement the judgmental assessments typical of official-sector bodies.

The visualization is a **heat map**: each indicator and each component is rendered as a color from blue (subdued) to red (acute), where every cell is the indicator's quantile in its own historical distribution. Colors are relative to history, not absolute levels. The paper's headline empirical claims: the summary vulnerability measures pick up rising imbalances through the mid-2000s, presaging the financial crisis, and they lead the credit-to-GDP gap (a key Basel III gauge) by a year or more, making them potentially useful for setting macroprudential tools such as the countercyclical capital buffer.

The 14 components and their indicators are:

1. **Equity markets:** E/P minus 10-year Treasury yield; E/P ratio.
2. **Price volatility:** VIX; CDS premium.
3. **Business credit:** BBB spread; high-yield spread; junk plus leveraged-loan issuance; C&I lending standards.
4. **Commercial real estate:** CRE prices; change in CRE lending standards.
5. **Residential real estate:** FICO scores; house prices relative to rents; change in mortgage lending standards.
6. **Nonfinancial business:** real debt growth; net leverage of risky firms; deep-junk issuance share; interest expenses; debt-to-income ratio.
7. **Consumer credit:** consumer credit/GDP; consumer debt service ratio; riskier-borrower debt/DPI; rapid-borrowing incidence.
8. **Home mortgages:** mortgage debt/GDP; mortgage debt service ratio; riskier-borrower mortgage debt/DPI; piggy-back incidence; rapid-borrowing incidence.
9. **Net saving:** household saving/GDP; nonfinancial business saving/GDP.
10. **Bank leverage:** tangible common equity/assets; Tier 1 common equity ratio; total capital ratio.
11. **Nonbank leverage:** broker-dealer financing; broker-dealer leverage ratio; non-agency securitization/GDP.
12. **Maturity transformation:** loans/deposits; weighted-average maturity gap; net short-term wholesale debt/GDP.
13. **Short-term wholesale funding:** gross short-term wholesale debt/GDP; short-term money share of BHC assets; runnable money-like liabilities/GDP.
14. **Size/concentration:** financial-sector liabilities/GDP; top-5 BHC asset share; illiquidity concentration.

2 The index construction

Every indicator passes through an identical pipeline. Let $x_{i,t}$ be the raw quarterly series for indicator i, t indexing quarters. The estimation sample is 1990Q1–2014Q4 (100 quarters); series with shorter histories use their full available span at every step.

Step 1 – sign orientation. Each series is multiplied by $s_i \in \{+1, -1\}$ so that higher values mean greater vulnerability:

$$y_{i,t} = s_i \cdot T(x_{i,t}). \tag{1}$$

Spreads and volatility enter with $s_i = -1$ on their logged values: compressed spreads and low volatility read as *complacency*, which the paper treats as the vulnerable state. Capital ratios enter with -1 (higher capital = safer); saving rates with -1 (higher saving = less vulnerable); lending standards with -1 (net tightening = safer). The exact sign per indicator is listed in Table 1.

Step 2 – transform T. One of three transforms:

- `log`: $T(x_t) = \ln x_t$ (natural log; strictly positive, right-skewed series: VIX, CDS premium, BBB and high-yield spreads, junk/leveraged-loan issuance);
- `ma10`: $T(x_t) = x_t - \bar{x}_t^{(40)}$, where $\bar{x}_t^{(40)}$ is the trailing 40-quarter moving average computed with a minimum of 20 observations (the first up to 19 quarters are missing). Used for ratio-to-GDP series and other trending levels, so the index responds to the *cyclical* component;
- `none`: $T(x_t) = x_t$.

Step 3 – standardize.

$$z_{i,t} = \frac{y_{i,t} - \mu_i}{\sigma_i}, \tag{2}$$

with μ_i, σ_i the sample mean and sample standard deviation (denominator $n-1$) over the indicator's full available sample.

Step 4 – component index. Simple mean of the standardized indicators in component j:

$$c_{j,t} = \frac{1}{|J_j|} \sum_{i \in J_j} z_{i,t}. \tag{3}$$

Step 5 – kernel CDF. Each component is mapped to $(0, 1)$ through a Gaussian kernel CDF with Silverman bandwidth, evaluated at each observation:

$$\hat{F}_j(c) = \frac{1}{n} \sum_{s=1}^{n} \Phi\left(\frac{c - c_{j,s}}{h_j} \right), \qquad h_j = 1.06\, \hat{\sigma}_j\, n^{-1/5}, \tag{4}$$

where Φ is the standard normal CDF, $\hat{\sigma}_j$ the sample standard deviation of the component series, and n the number of non-missing observations. This is what makes each heat-map cell a historical quantile.

Step 6 – aggregate index. Mean of the 14 components on $(0, 1)$, then mapped through its own kernel CDF again:

$$a_t = \frac{1}{14} \sum_{j=1}^{14} \hat{F}_j(c_{j,t}), \qquad A_t = \hat{F}_a(a_t). \tag{5}$$

Table 1 gives the component, sign, and transform for all 44 indicators, recovered from the paper's documentation and verified against its published indices (§6.1).

Table 1: Per-indicator specification: component, sign s_i, transform.

Indicator	Component	s_i	Transform
E/P minus 10-year Treasury yield	Equity markets	-1	none
E/P ratio	Equity markets	-1	none
VIX	Price volatility	-1	log
CDS premium	Price volatility	-1	log
BBB spread	Business credit	-1	log
High-yield spread	Business credit	-1	log
Junk + leveraged-loan issuance	Business credit	$+1$	log
C&I lending standards	Business credit	-1	none
CRE prices	Commercial real estate	$+1$	ma10
Change in CRE lending standards	Commercial real estate	-1	none
FICO scores	Residential real estate	-1	none
House prices relative to rents	Residential real estate	$+1$	ma10
Change in mortgage lending standards	Residential real estate	-1	none
Real debt growth, nonfinancial business	Nonfinancial Business	$+1$	none
Net leverage of risky firms	Nonfinancial Business	$+1$	none
Deep-junk issuance share	Nonfinancial Business	$+1$	none
Interest expenses	Nonfinancial Business	$+1$	none
Debt-to-income ratio	Nonfinancial Business	$+1$	ma10
Consumer credit / GDP	Consumer credit	$+1$	ma10
Consumer debt service ratio	Consumer credit	$+1$	none
Riskier-borrower debt / DPI	Consumer credit	$+1$	none
Rapid-borrowing incidence	Consumer credit	$+1$	none
Mortgage debt / GDP	Home mortgages	$+1$	ma10
Mortgage debt service ratio	Home mortgages	$+1$	none
Riskier-borrower mortgage debt / DPI	Home mortgages	$+1$	none
Piggy-back incidence	Home mortgages	$+1$	none
Rapid-borrowing incidence	Home mortgages	$+1$	none
Household saving / GDP	Net saving	-1	none
Nonfinancial business saving / GDP	Net saving	-1	none
Tangible common equity / assets	Bank leverage	-1	none
Tier 1 common equity ratio	Bank leverage	-1	none
Total capital ratio	Bank leverage	-1	none
Broker-dealer financing	Nonbank leverage	$+1$	none
Broker-dealer leverage ratio	Nonbank leverage	-1	none
Non-agency securitization / GDP	Nonbank leverage	$+1$	none
Loans / deposits	Maturity transformation	$+1$	none
Weighted-average maturity gap	Maturity transformation	$+1$	none
Net short-term wholesale debt / GDP	Maturity transformation	$+1$	ma10
Gross short-term wholesale debt / GDP	Short-term wholesale funding	$+1$	ma10
Short-term money share of BHC assets	Short-term wholesale funding	$+1$	none
Runnable money-like liabilities / GDP	Short-term wholesale funding	$+1$	ma10
Financial-sector liabilities / GDP	Size/concentration	$+1$	ma10
Top-5 BHC asset share	Size/concentration	$+1$	none

Table 1: (continued)

Indicator	Component	s_i	Transform
Illiquidity concentration	Size/concentration	+1	none

3 Ground truth: the accessible materials

The paper's page links "Accessible materials (.zip)", downloadable at

`https://www.federalreserve.gov/econresdata/feds/2015/files/feds2015059.zip`

It contains HTML tables, of which two matter for replication:

- **The raw-indicators table:** 44 series, quarterly, 1990Q1–2014Q4 (100 quarters; shorter histories for series that start later, e.g. CDS premia from 2005Q2, the FICO series from 2003Q1).
- **The indices table:** the published aggregate index plus the 14 component indices, already mapped to $(0, 1)$.

These tables are the ground truth for the entire exercise: every proxy is scored against the raw table, and every pipeline output is scored against the indices table. Downloading and parsing them is step 1 of the recipe (§8). Note the accessible column names carry the paper's own typos ("Volitility", "Ratro", "Conventration", "Commerical", "Mortage", "Comsumer", "Home Mortage")—match columns by position and concept, not by spelling.

4 Indicator catalog: public sources and constructions

For each of the 44 indicators: the public source with its exact identifier, the precise construction, and—where a proxy stands in for proprietary or microdata—what it proxies, why it was chosen, and what dynamics it captures. Correlations quoted are the proxy's correlation against the accessible series on the *transformed* signal (sign, transform, and standardization applied as in §2), over the stated overlap; because the pipeline standardizes everything, only the transformed dynamics matter for the heat map. A standing caveat for every series: current public vintages are *revised* data; the paper used 2015 vintages, so part of every deviation is data revision, not construction error.

Frequency and timing conventions used throughout: "quarterly mean" of a daily series is the arithmetic mean of daily observations in the calendar quarter; of a monthly series, the mean of the three months. Year-on-year percent change is $100\,(x_t/x_{t-4} - 1)$. FRED series are fetched from the public CSV endpoint `https://fred.stlouisfed.org/graph/fredgraph.csv?id=SERIES_ID` (replacing `SERIES_ID`); the FRED server stalls non-browser user agents, so download with `curl` (default user agent), not Python `urllib`.

4.1 Equity markets

Earnings/Price minus 10-year Treasury yield. *Replicated.* Robert Shiller's `ie_data.xls` (`http://www.econ.yale.edu/~shiller/data/ie_data.xls`), E/P column, quarterly mean, minus FRED `DGS10` (10-year Treasury constant maturity yield, quarterly mean of daily), in percentage points. Transformed correlation vs. accessible: 0.85 (25 overlapping quarters; the accessible E/P columns start later).

Earnings/Price ratio. *Replicated.* Shiller E/P, quarterly mean. Transformed correlation 0.87. (See §5 for the orientation correction: the accessible column is E/P even though the data appendix discusses the P/E concept.)

4.2 Price volatility

VIX. *Replicated.* FRED `VIXCLS` (CBOE Volatility Index, close), quarterly mean; the natural log is applied at the index stage per Table 1. Correlation 1.00 (100 quarters).

CDS premium. *Proxy, 0.96 (39 quarters, 2005Q2–2014Q4).* The paper uses Markit investment-grade CDS premia (proprietary). Public analog: the monthly Moody's Baa credit spread (FRED `BAA` minus `DGS10`, in percentage points) *plus* the monthly Gilchrist–Zakrajšek excess bond premium (EBP), averaged to quarterly. EBP isolates the credit-risk-premium component of spreads and is downloaded from the Federal Reserve's permanent CSV at `https://www.federalreserve.gov/econres/notes/feds-notes/ebp_csv.csv` (columns `date`, `gz_spread`, `ebp`, `est_prob`; monthly from 1973M1). Rationale: CDS premia are the market price of investment-grade credit risk; the Baa spread is the standard public investment-grade credit-spread gauge, and EBP adds the sentiment-driven premium component. Rejected alternatives: the GZ total credit spread (0.90), a level-shifted EBP (0.82—the shift is arbitrary and EBP takes negative values, which breaks the pipeline's log transform).

4.3 Business credit

BBB spread. *Proxy, 0.96 (100 quarters).* FRED `BAA` (Moody's Baa corporate bond yield) minus `DGS10`, quarterly mean, in percentage points. The paper uses the ICE BofA BBB option-adjusted spread in basis points; the Moody's Baa spread is the public analog (units differ—bp vs. pp—which washes out in standardization).

High-yield spread. *Proxy, 0.86 (100 quarters).* Monthly Moody's Baa–Treasury spread plus monthly EBP (same Fed CSV as above), averaged to quarterly. Economic logic: HY OAS equals the investment-grade credit spread plus extra compensation for higher expected defaults and lower liquidity; EBP isolates the credit-risk premium, so adding it to the Baa spread captures HY-specific dynamics the Baa spread alone misses ($0.72 \rightarrow 0.86$). Direct use of ICE BofA HY OAS is impossible: FRED's public CSV for that series is truncated to about three years of history (redistribution restrictions).

Junk plus leveraged-loan issuance (investor appetite). *Proxy, 0.82 (52 quarters, 2002Q1–2014Q4).* The paper uses HY bond plus leveraged-loan issuance (Dealogic/LCD, proprietary; SIFMA's public quarterly history is truncated to recent quarters in every archived vintage; no public quarterly leveraged-loan issuance series exists). Proxy: the *inverse* average corporate credit spread, 1/(GZ spread in percent), quarterly mean of monthly—when spreads are tight, investor appetite for risky debt is strong and issuance surges, a standard risk-appetite gauge. The GZ spread is the `gz_spread` column of the Fed EBP CSV above. Rejected: 1/VIX (0.82), 1/Baa spread (0.59), Z.1 net corporate bond issuance (0.28—IG issuance surged in 2008–09 while HY collapsed, the wrong dynamics), 1/expected default probability (0.67).

C&I lending standards. *Replicated.* FRED `DRTSCILM` (net percent of domestic banks tightening standards for C&I loans to large and middle-market firms), quarterly. Correlation 0.90 (99 quarters).

4.4 Commercial real estate

CRE prices. *Proxy, 0.98 (81 quarters).* Bank for International Settlements quarterly commercial property price index for the United States, nominal, as published (index level; the BIS series runs 1945Q4–2026Q2). The paper applies the 10-year-MA detrend (Table 1); level and base differences wash out in the transform.

Change in CRE lending standards. *Proxy, 1.00 (96 quarters).* Net percent of domestic banks tightening standards for CRE loans, from the Senior Loan Officer Opinion Survey (SLOOS) historical XML published by the Federal Reserve Board. Two splices: 1990Q2–2013Q2 uses the discontinued aggregate series `SUBLPDRCS_N.Q` ("all commercial real estate loans"), re-dated back one quarter to match the paper's reference-quarter convention (the paper dates each survey to the quarter it asks about; correlation 1.00 over 90 quarters after re-dating). 2013Q3–2014Q4 splices the mean of the three successor sub-questions—construction and land development `SUBLPDRCSC_N.Q`, nonfarm nonresidential `SUBLPDRCSN_N.Q`, multifamily `SUBLPDRCSM_N.Q`—with the same dating. The paper's exact post-split splice was not reverse-engineered.

4.5 Residential real estate

FICO scores. *Proxy, 0.99 (48 quarters, 2003Q1–2014Q4).* Annual UPB-weighted mean origination FICO for GSE (Fannie Mae) single-family acquisitions, from Fannie Mae's credit supplements, cubic-spline interpolated from annual (mid-year) to quarterly. The concept matches the paper's: "FICO scores for mortgages sold to GSEs."

House prices relative to rents. *Replicated.* FRED `CSUSHPISA` (Case-Shiller national home price index) divided by `CUSR0000SEHA` (rent of primary residence CPI). Correlation 1.00 (100 quarters).

Change in mortgage lending standards. *Proxy, 0.91 (31 quarters, 2007Q2–2014Q4).* FRED `DRTSPM` (net percent tightening for prime residential mortgages). Caveat: the paper uses all-residential SLOOS from 1990; `DRTSPM` covers prime mortgages only, from 2007Q2.

4.6 Nonfinancial business

Real debt growth. *Proxy, 0.80 (96 quarters).* Year-on-year percent change of GDP-deflated total nonfinancial *business* debt (corporate plus noncorporate; FRED `TCMILBSNNB`—total credit market instruments, liabilities, nonfinancial business—deflated by `GDPDEF`). The paper covers all nonfinancial business (corporate + noncorporate); a corporate-only construction scores only 0.31.

Net leverage of risky firms. *Proxy, 0.80 (100 quarters).* 100×(nonfinancial corporate business debt securitie quarterly NSA levels: FRED `BCNSDODNS` (NFCB debt securities and loans, liability, $M) over FRED `BOGZ1FL104090103Q` (NFCB total assets, $M)—an aggregate net-leverage analogue of the paper's Compustat-based net leverage of speculative-grade/unrated firms. Rejected: a net-of-checkable-deposits variant (0.74), EBP (0.24).

Deep-junk issuance share. *Proxy, 0.37—honest but weak, kept in the run (87 quarters, 1993Q1–2014Q4).* The paper uses the CCC-or-below share of high-yield issuance; no public quarterly series exists (SIFMA does not break HY out by rating; deal databases are proprietary). Proxy: negated excess bond premium—when credit sentiment is strong (low/negative EBP), low-quality firms can issue and the deep-junk share rises. Raw = −EBP in percent, quarterly mean of monthly. Rejected: HY/Baa/total credit spreads as coincident proxies (−0.16 to 0.08), −VIX (0.11), −expected default probability (0.35), −SLOOS C&I standards (0.30). Sensitivity: excluding it lifts the Nonfinancial Business component from 0.62 to 0.67 and the aggregate from 0.931 to 0.935.

Interest expenses (debt burden). *Proxy, 0.72 (100 quarters).* Quarterly, seasonally adjusted, from the BEA National Income and Product Accounts bulk file (`https://apps.bea.gov/national/Release/TXT/NipaDataQ.txt`), Table 1.14 (GDI):

$$100 \times \frac{\texttt{B471RC}}{\texttt{A466RC} - \texttt{B467RC} + (\texttt{A455RC} - \texttt{A457RC})}, \tag{6}$$

where `B471RC` is net interest and miscellaneous payments (nonfinancial corporate), `A466RC` − `B467RC` is undistributed after-tax profits excluding IVA/CCAdj (profits after tax without inventory valuation and capital consumption adjustments, minus net dividends), and `A455RC` − `A457RC` is consumption of fixed capital (gross minus net value added). Cash flow is retained profits plus economic depreciation; excluding IVA/CCAdj keeps the denominator on a cash-like basis.

Debt-to-income ratio. *Proxy, 0.76 (81 quarters).* 100×(corporate+noncorporate debt securities and loans)/ in percent (FRED `TCMILBSNNB` / `GDP`).

4.7 Consumer credit

Consumer credit / GDP. *Replicated.* FRED `TOTALSL` (total consumer credit outstanding) / `GDP`. Correlation 1.00 (100 quarters).

Consumer debt service ratio. *Proxy, 0.81 (40 quarters, 2005Q1–2014Q4).* FRED `TDSP` (total household debt service ratio). Caveat: the paper uses a consumer-only DSR; `TDSP` covers all household debt including mortgages.

Riskier-borrower debt / DPI. *Proxy, 0.83 (64 quarters).* Quarterly average of monthly revolving consumer credit outstanding (FRED `REVOLSL`, G.19) divided by quarterly average nominal DPI (BEA via FRED). Revolving (credit-card) balances are the consumer-credit segment where riskier borrowers concentrate; the stock/DPI ratio tracks the paper's FRBNY CCP-microdata-based riskier-borrower debt stock.

Rapid-borrowing incidence. *Proxy, 0.82 (60 quarters).* Year-on-year percent change of quarterly revolving consumer credit outstanding (`REVOLSL`). When aggregate revolving balances grow rapidly, a larger share of (riskier) borrowers are borrowing rapidly, tracking the paper's CCP-based incidence of very rapid borrowing.

4.8 Home mortgages

Mortgage debt / GDP. *Replicated.* FRED `HHMSDODNS` (households, 1–4 family mortgage debt) / `GDP`. Correlation 1.00 (100 quarters).

Mortgage debt service ratio. *Replicated.* FRED `MDSP`. Correlation 0.99 (40 quarters).

Riskier-borrower mortgage debt / DPI. *Proxy, 0.87 (64 quarters).* Quarterly household mortgage debt (FRED `HHMSDODNS`, converting $M to the $B units of DPI) divided by nominal DPI, in percent. The riskier-borrower share of mortgage debt is procyclical, so the aggregate mortgage-debt stock captures the paper's CCP-based riskier-borrower debt/DPI dynamics.

Piggy-back incidence. *Proxy, 0.74 (63 quarters).* Year-on-year percent change of quarterly household mortgage debt (`HHMSDODNS`). Piggy-back (subordinate-lien-at-purchase) originations concentrate in mortgage-credit booms; the paper's CCP-based incidence series tracks aggregate mortgage growth. A direct subordinate-lien share would need HMDA loan-level data (annual, and the 1999–2006 national files are not retrievable), hence the quarterly public proxy.

Rapid-borrowing incidence. *Proxy, 0.96 (60 quarters).* Year-on-year percent change of quarterly household mortgage debt—when aggregate mortgage balances grow rapidly, a larger share of borrowers show very rapid balance growth.

4.9 Net saving

Household saving / GDP. *Proxy, 0.82 (100 quarters).* FRED PSAVERT × DSPI / GDP. PSAVERT is percent of DPI; converting through DSPI/GDP puts it on the GDP ratio the paper uses.

Nonfinancial business saving / GDP. *Proxy, 0.84 (100 quarters).* Quarterly NIPA net lending (+) / net borrowing (−) of domestic business—net saving net of capital formation, the paper's "savings net of capital formation" concept—divided by GDP, in percent: BEA NIPA bulk file Table 5.1, line 37 W995RC (net lending, domestic business, SA) over line 1 A191RC (GDP, SA). Caveat: current-vintage Table 5.1 no longer splits financial vs. nonfinancial business; domestic-business net lending is the closest public quarterly aggregate.

4.10 Bank leverage

Tangible common equity / tangible assets. *Proxy, 0.90 (100 quarters).* Quarterly aggregate across all FR Y-9C filers (Federal Reserve Bank of Chicago bulk BHCF files):

$$100 \times \frac{\texttt{BHCK3217} - \texttt{BHCK3163}}{\texttt{BHCK2170} - \texttt{BHCK3163}}, \tag{7}$$

where BHCK3217 is total equity capital, BHCK3163 intangible assets, and BHCK2170 total assets. Matches the paper's BHC tangible-leverage concept; replaces a commercial-bank EQTA proxy (0.36 → 0.90).

Tier 1 common equity ratio. *Proxy, 1.00 (52 quarters, 2001Q1–2013Q4).*

$$100 \times \frac{\texttt{BHCK8274} - \texttt{BHCK3283} - \texttt{BHCK3000} - \texttt{BHCKC502}}{\texttt{BHCKB704}}, \tag{8}$$

aggregated across all Y-9C filers: Tier 1 capital (BHCK8274) minus perpetual preferred stock (BHCK3283, HC-23) minus minority interest (BHCK3000) minus qualifying trust preferred securities (BHCKC502, HC-R memorandum item, available from 2004Q1), over total risk-weighted assets (BHCKB704, HC-R). Mnemonics validated against Federal Reserve edit rules. Coverage is 2001Q1–2013Q4: 2014Q1–Q4 are excluded because Basel III revised the Y-9C HC-R in 2014Q1 (Tier 1 became BHCA8274 with new risk-weighted-asset definitions), breaking comparability—the accessible series shows no break, indicating a different 2014 methodology. Pre-2004 trust preferred (C502) is unavailable and treated as zero (small then).

Total capital ratio. *Proxy, 0.99 (16 quarters, 1996Q1–2010Q4, sparse).*

$$100 \times \frac{\texttt{RCFD8274} + \texttt{RCFD8275}}{\texttt{RCFDA223}}, \tag{9}$$

aggregated across all Call Report filers (FFIEC Central Data Repository bulk downloads): Tier 1 (RCFD8274) plus Tier 2 (RCFD8275) over net risk-weighted assets (RCFDA223, Schedule RC-R), using consolidated (RCFD) items to cover both FFIEC 031 and 041 filers. Validated: 2010Q4 proxy 15.12% vs. accessible 15.09%—a near-exact level match. Overlap is thin (16 quarters) because the bulk download had limited Call Report quarters (no 1990Q1/1993Q1 RC-R items; no post-2010Q4 files retrieved).

4.11 Nonbank leverage

Broker-dealer financing. *Proxy, 0.94 (54 quarters, accessible from 2001Q3).*

$$100 \times \frac{\texttt{FL662051003.Q} + \texttt{FL662151003.Q}}{\text{GDP}}, \tag{10}$$

Z.1 Financial Accounts mnemonics: security repurchase agreement assets (FL662051003.Q; reverse repos, financing provided) plus security repurchase agreement liabilities (FL662151003.Q) of security brokers and dealers. Gross (both sides) securities financing; the level match is near-exact (2008Q1: proxy 32.6 vs. accessible 32.6). Z.1 series are available on FRED under BOGZ1-prefixed codes.

Broker-dealer leverage ratio. *Proxy, 0.61 (100 quarters)—moderate, see §5.*

$$100 \times \frac{\texttt{FL664090005.Q} - \texttt{FL664190005.Q}}{\texttt{FL664090005.Q}}, \tag{11}$$

i.e. equity over assets for security brokers and dealers (FRED equivalents BOGZ1FL664090005Q, BOGZ1FL664190005Q). Subperiod correlations: 1990–1999 0.88, 2007–2009 0.92, 2010–2014 ≈ 0.0—post-crisis Z.1 vintage revisions to broker-dealer balance sheets changed dynamics relative to the 2015-vintage data the paper used. Crisis dynamics are captured well.

Non-agency securitization / GDP. *Proxy, 0.94 (49 quarters, accessible from 2002Q4).* Z.1 ABS-issuer sector net transactions / GDP, in percent: FRED BOGZ1FA674190005Q ($M SAAR) divided by 1000 and by GDP ($B SAAR), times 100. By construction the ABS-issuer sector excludes agency/GSE pools, so this is the quarterly public net-issuance measure of the same private-label (non-agency MBS + ABS) universe the paper measures with SIFMA gross issuance. Rejected: SIFMA ABS/MBS issuance (public quarterly history covers only recent quarters in all archived vintages; annual only in older files).

4.12 Maturity transformation

Loans / deposits. *Proxy, 0.97 (73 quarters, accessible from 1996Q4).*

$$100 \times \frac{\texttt{BHCK2122}}{\texttt{BHDM6631} + \texttt{BHDM6636} + \texttt{BHFN6631} + \texttt{BHFN6636}}, \tag{12}$$

aggregated across all Y-9C filers: total loans and leases (BHCK2122, Schedule HC-C, gross) over deposits—domestic offices (BHDM6631+BHDM6636) plus foreign offices (BHFN6631+BHFN6636) from Schedule HC-E. The proxy level runs ≈18–20pp below the accessible (a consistent gap, likely a narrower deposit definition or different loan aggregate in the paper); shape and dynamics match well, and the level gap washes out in standardization.

Weighted-average maturity gap. *No workable public proxy*—the one failure; see §6.3.

Net short-term wholesale debt / GDP. *Proxy, 0.94 (81 quarters).*

$$100 \times \frac{\texttt{FL794141005.Q} + \texttt{FL793169175.Q} - \texttt{FL792050005.Q} - \texttt{FL793069175.Q}}{\text{GDP}}, \tag{13}$$

short-term loan liabilities plus open market paper liabilities minus federal funds & repo assets minus open market paper assets, domestic financial sectors, Z.1. Netting removes the financial sector's role as a wholesale lender; 2008Q1 level: proxy 23.5 vs. accessible 24.0.

4.13 Short-term wholesale funding

Gross short-term wholesale debt / GDP. *Proxy, 0.95 (81 quarters).*

$$100 \times \frac{\texttt{FL794141005.Q} + \texttt{FL793169175.Q}}{\text{GDP}}. \tag{14}$$

"Short-term loans" in Z.1 comprises federal funds plus repos plus other short-term borrowing (verified: short-term loans ≈ repos + other loans and advances, corr 0.99)—the short-term wholesale (noncore liabilities) concept.

Short-term money share of BHC assets. *Proxy, 0.90 (56 quarters).*

$$100 \times \frac{\text{fed funds purchased} + \text{repos} + \text{other borrowed money}_{\leq 1\text{yr}}}{\texttt{BHCK2170}}, \tag{15}$$

aggregated across all Y-9C filers. Y-9C mnemonics: fed funds purchased plus securities sold under agreements to repurchase = `BHCK2800` (1990Q1–2001Q4), spliced to `BHCKB995`+`BHCKB996` (2002Q1 onward; splice verified smooth: 9.23% → 8.84% of assets); other borrowed money with remaining maturity ≤ 1 year = `BHCK2332`; total assets = `BHCK2170`. Excludes other borrowed money > 1yr (`BHCK2333`)—including it lowers correlation. The proxy level (≈15% of assets) runs well below the accessible (≈21–38%): the paper's "short-term money" is a broader aggregate than fed funds + repos + short-term borrowings; the exact definition could not be pinned down from public Y-9C items. Dynamics (pre-crisis rise, post-crisis fall) are captured.

Runnable money-like liabilities / GDP. *Proxy, 0.95 (81 quarters).*

$$\frac{\texttt{FL792150005.Q} + \texttt{FL793169175.Q} + \texttt{FL634090005.Q}}{\text{GDP}}, \tag{16}$$

federal funds & security repo liabilities plus open market paper liabilities plus money market fund shares outstanding (money market funds total financial assets = shares outstanding), domestic financial sectors, Z.1. Omits uninsured deposits, variable-rate demand obligations, and securities lending (not in Z.1); the residual is smooth and slow-moving, so cyclical dynamics are captured.

4.14 Size / concentration

Financial-sector liabilities / GDP. *Replicated.* FRED `BOGZ1FL794190005Q` (Z.1 domestic financial sectors liabilities) / `GDP`. Correlation 0.99 (100 quarters).

Top-5 BHC asset share. *Proxy, 0.98 (100 quarters).*

$$100 \times \frac{\sum_{i \in \text{top 5 by BHCK2170}} \texttt{BHCK2170}_i}{\sum_{\text{all Y-9C filers}} \texttt{BHCK2170}}, \tag{17}$$

ranked within-quarter on consolidated total assets. Caveat: the bulk files mix filer universes across quarters (Q1/Q3 ≈ 1–2.6k filers vs. Q2/Q4 ≈ 5.7–5.9k filers, the latter including lower-tier/small BHCs with double-counting) and contain RSSD-change duplicates (identical assets under two RSSDs); both distort the denominator level in some quarters (notably 2005), though share dynamics track well. A top-tier-only filter would improve it, but the files carry no tier flag.

Illiquidity concentration. *Proxy, 0.90 (76 quarters, accessible from 1996Q1).*

$$\frac{\texttt{FL793063005.Q} + \texttt{FL794035005.Q}}{\texttt{FL794090005.Q}}, \tag{18}$$

corporate and foreign bond assets plus total loan assets, over total financial assets of domestic financial sectors, Z.1. This proxies the paper's illiquidity-concentration framework (a modified Herfindahl of illiquid asset holdings from the fire-sale spillovers literature): as illiquid credit assets grow as a share of financial-sector balance sheets, fire-sale spillover vulnerability rises. Rejected: the Treasury-market "noise" measure—the authors' published data was unfetchable, and it has the wrong dynamics (noise is a flow/market-price measure spiking in 2008; the target is a stock/balance-sheet measure peaking pre-crisis in 2006 and falling through 2008).

5 Two empirical orientation corrections

The paper's data appendix describes each indicator's concept, but two accessible columns did not behave as described. Both were found the same way: correlating the accessible sub-series against the published component index and finding the sign wrong. A reader following this guide will hit the same finding at the same step.

E/P vs. P/E. The appendix discusses equity valuations in price/earnings terms, but the accessible "Earnings/Price ratio" column is the *inverse* of that concept—it is E/P (higher = cheaper = less vulnerable). Entering it with the appendix-implied sign gave a component correlation of -0.96; flipping the sign (the series enters with sign -1, as does E/P minus the 10-year yield) gives $+0.96$.

Broker-dealer leverage as a capital ratio. The appendix's "broker dealer leverage ratio" reads as a leverage concept (higher = more vulnerable), but the accessible series behaves as a *capital* ratio: mean 2.4%, minimum 0.7% in 2008Q3, maximum 4.3% in 1990Q2—equity over assets, where higher means safer. Entering it with the appendix-implied sign gave a component correlation of -0.17; building the proxy as equity/assets with sign -1 gives $+0.93$. (The companion "dealer financing" series needed no correction.)

6 Replication results

6.1 Pipeline check: the documented transforms on the accessible data

Applying the §2 pipeline to the *accessible raw series* and comparing against the *published* component indices:

Table 2: Pipeline check: documented transforms applied to the accessible raw series vs. published indices.

Component	n	Correlation	Max abs. dev.
Equity markets	100	0.9571	0.2383
Price volatility	100	0.8807	0.4254
Business credit	100	0.9095	0.5613
Commercial real estate	99	0.6215	0.6072
Residential real estate	98	0.8581	0.4538
Nonfinancial Business	100	0.8511	0.4451
Consumer credit	100	0.9823	0.1241
Home mortgages	100	0.9875	0.1259
Net saving	100	0.9588	0.2571
Bank leverage	100	0.9822	0.2131
Nonbank leverage	100	0.9312	0.5810
Maturity transformation	81	0.9624	0.3385
Short-term wholesale funding	81	0.9730	0.3118
Size/concentration	100	0.9796	0.1628
Aggregate index	100	**0.9693**	0.2806

13 of 14 components ≥ 0.85; aggregate 0.97. The heat-map figure rebuilt from these components matches the paper's. The method, in other words, is fully reproducible from its own documentation—except for commercial real estate (see §6.3).

6.2 End-to-end: the full public-proxy panel through the pipeline

Running the identical pipeline on all 43 public proxies (masked to accessible raw coverage so component composition matches the published construction), compared against the published indices:

Table 3: End-to-end: full public-proxy panel through the pipeline vs. published indices.

Component	n	Correlation	Max abs. dev.
Equity markets	25	0.8811	0.4161
Price volatility	100	0.8733	0.4027
Business credit	100	0.8192	0.5219
Commercial real estate	99	0.5789	0.6132
Residential real estate	81	0.8856	0.4571
Nonfinancial Business	100	0.6241	0.6474
Consumer credit	81	0.9025	0.3396
Home mortgages	81	0.9861	0.1846
Net saving	100	0.8324	0.5009
Bank leverage	100	0.9056	0.3261
Nonbank leverage	100	0.8614	0.5000
Maturity transformation	81	0.9193	0.2839
Short-term wholesale funding	81	0.9568	0.3798
Size/concentration	100	0.9509	0.2295
Aggregate index	100	**0.9311**	0.4004

10 of 14 components ≥ 0.85; aggregate 0.93. (Excluding the weak deep-junk proxy: Nonfinancial Business 0.67, aggregate 0.935.)

6.3 The honest misses

Commercial real estate (0.58) is a pipeline discrepancy, not a proxy failure. The documented pipeline reproduces only 0.62 *even on the accessible raw series* (Table 2)—and both CRE proxies themselves score ≥ 0.98. Something in the paper's CRE construction differs from its documentation (likely the standards splice or the detrending window); the proxy panel cannot fix what the accessible data cannot reproduce. This is a finding about the paper.

Nonfinancial Business (0.62)—the deep-junk issuance share proxy (negated EBP, 0.37) is honest but weak; no public quarterly CCC-share series exists. Interest expenses (0.72, NIPA corporate only) and debt/income (0.76) are mediocre.

Business credit (0.82)—the HY spread via Baa+EBP (0.86) and issuance via inverse GZ spread (0.82) capture most but not all HY dynamics.

Net saving (0.83)—household (0.82) and business (0.84) saving proxies are close; marginal.

The maturity gap has no workable public proxy. The weighted-average maturity gap at commercial banks (the Brunnermeier–Oehmke concept) cannot be built from public data: the FFIEC CDR bulk Call Report files carry no descriptive labels (MDRM codes only, e.g. `RCON8693`–`RCON8716`), the FFIEC form documentation mapping MDRMs to the RC-C/RC-E maturity-bucket schedules was not in the downloaded data, and the accessible gap is essentially uncorrelated with loans/deposits (0.01 over 71 quarters)—confirming it cannot be proxied by simpler balance-sheet ratios and genuinely requires the maturity buckets.

7 Visual proof and extension to the present

7.1 Overlay charts: originals vs. proxies

Figures 1–4 overlay the accessible raw series (black) against the public proxy (red dashed) for 16 indicators spanning every vulnerability category, each standardized over the common sample with its correlation annotated. Standardization is the right comparison: the pipeline standardizes every input (§2), so co-movement of the standardized series is what the heat map consumes; raw levels and units are documented per series in §4. The clean replicates (VIX 1.000, mortgage debt/GDP 1.000, financial-sector liabilities/GDP 0.994, house prices/rents 0.995) are visually indistinguishable from the originals. The notable proxies track tightly: CRE prices via BIS (0.984), CRE standards via SLOOS (0.997), FICO via Fannie Mae (0.991), Tier 1 via FFIEC (0.90), BBB spread via Moody's Baa (0.955), high-yield spread via Baa plus EBP (0.897). Figure 4, panel 4, shows the honest weak proxy on the same scale as the successes: the negated-EBP deep-junk share (0.37) captures the crisis spike but misses the level and timing elsewhere—it is kept in the panel with its weakness disclosed, not hidden.

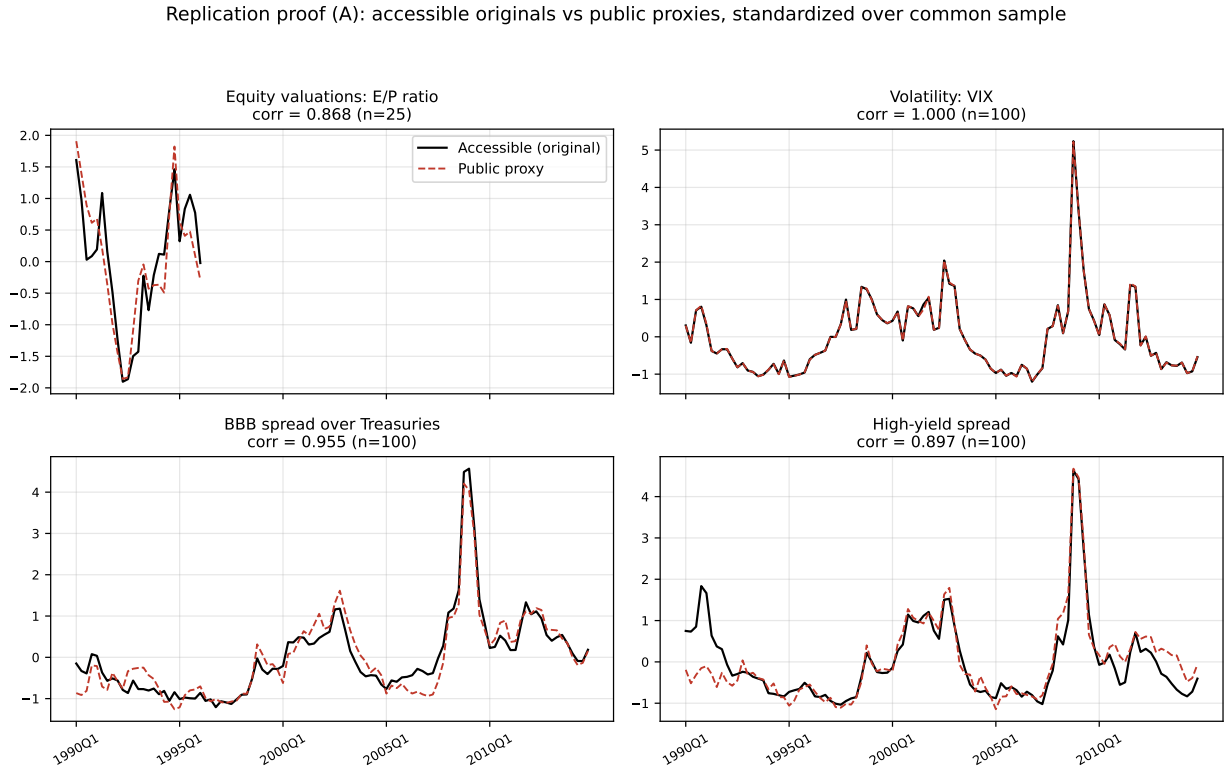


Figure 1: Replication proof (A): accessible originals vs. public proxies, valuations and volatility. Standardized over the common sample; correlation annotated per panel.

7.2 The heat map carried past 2015

The paper's sample ends 2014Q4, but the public proxies extend further: 40 of 43 series reach 2026Q2 or later (31 at 2026Q2, eight at 2026Q3, VIX at 2026Q4—a partial quarter), and the two E/P series stop at 2023Q1 (the Yale-hosted Shiller file has not been updated since September 2023).

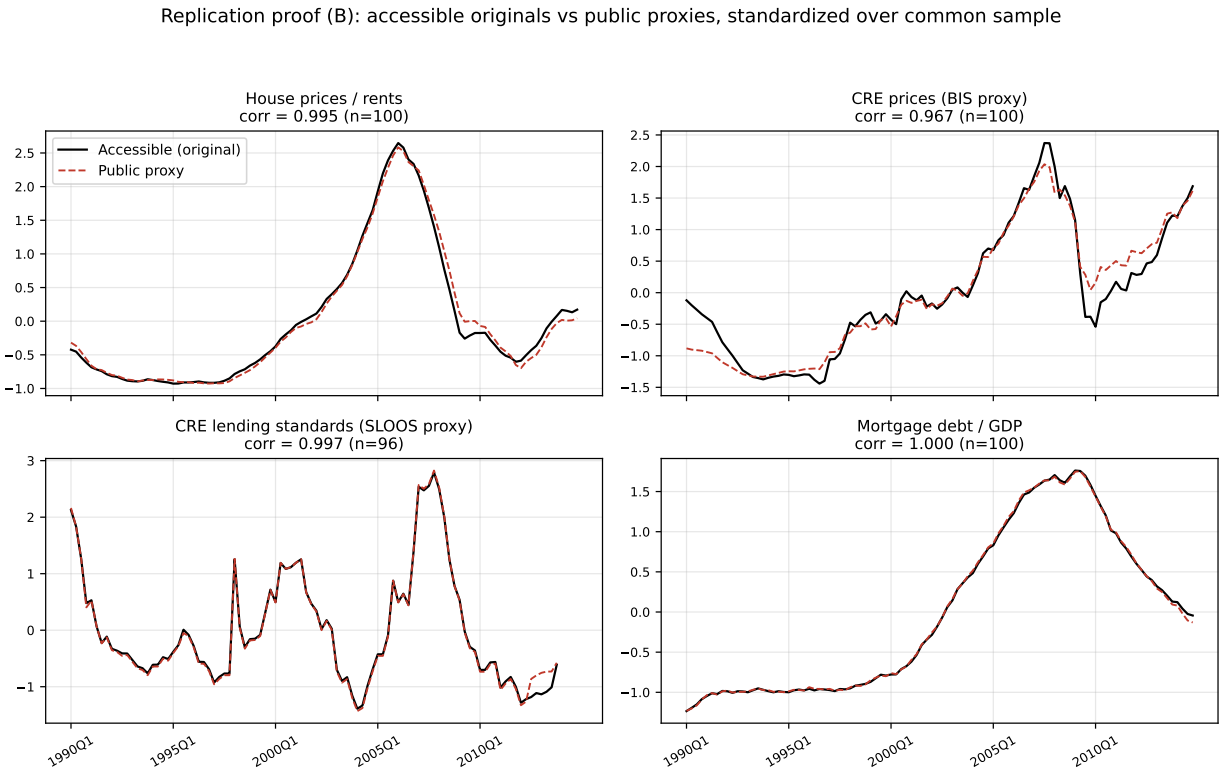


Figure 2: Replication proof (B): real estate and mortgages.

The prime-mortgage SLOOS series ends 2014Q4 (FRED discontinued it); FRED also discontinued the EQTA bank-leverage aggregate in 2020Q3, where the FFIEC-built tangible series takes over. Running the identical pipeline on the unmasked 1990Q1–2026Q4 proxy panel gives the heat map in Figure 5, the aggregate index in Figure 6, and all 14 components in Figure 7.

Three readings, stated plainly. **First, the extension validates the instrument:** the pre-2015 portion reproduces the paper's story—heat peaks at 0.98 in 2006Q4–2007Q2 and unwinds through the crisis, exactly the vulnerability-then-realization pattern the paper documents. **Second, the COVID shock barely registers** (aggregate 0.19 in 2020Q1, 0.06 in 2020Q2): the heat map measures complacency and imbalances, not stress, and the pandemic was an exogenous shock with no prior buildup—note the sign convention (§2), under which the VIX spike itself reads as *cool*. **Third, the recent picture is cool to lukewarm:** the aggregate sits at 0.18–0.26 through 2024–2026Q2, with warmth concentrated in business credit (0.80–0.89 over the last year) and price volatility (0.57–0.81); household, leverage, and funding components are all cool.

7.3 Caveats specific to the extension

- **History moves when the sample grows.** Standardization, the 10-year moving-average detrending, and the kernel-CDF bandwidth are all computed over the full 1990–2026 sample, so the 1990–2014 component values differ from both the published and the masked runs (aggregate correlation 0.96, max absolute deviation 0.36; worst component size/concentration, 0.59). The extended map is the same methodology on a longer sample—it is not point-for-point comparable with the paper's map.
- **The equity-markets component is missing from 2023Q1,** when the Shiller source

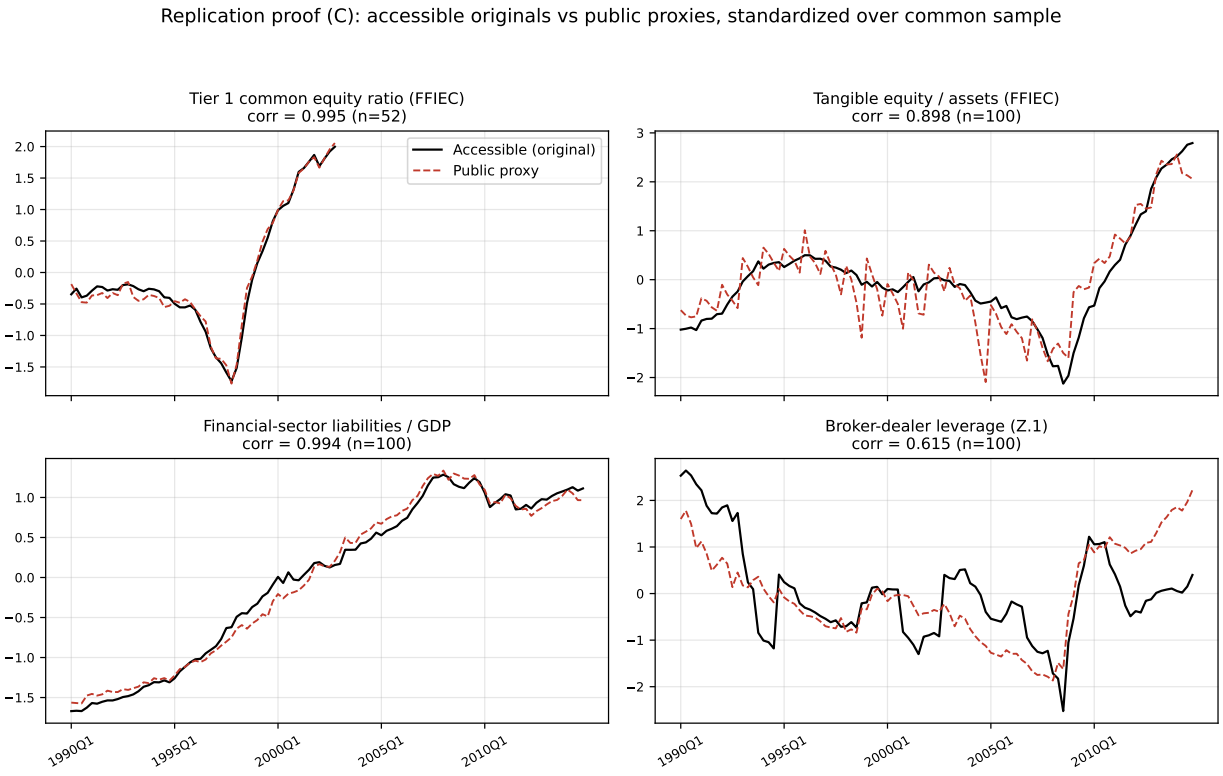


Figure 3: Replication proof (C): leverage—bank, nonbank, and financial-sector totals.

froze; the aggregate from that quarter is a 13-component mean. 2026Q3–Q4 are ragged-edge quarters with several series not yet published.

- **Vintages.** Everything is current-vintage public data against the paper's 2015 vintages; revisions explain part of every deviation.
- The CRE pipeline discrepancy (§6.3) and the missing maturity gap carry over unchanged.

8 Reproduction recipe

From an empty directory to the reproduced heat map. No code from any repository is needed—only the sources named below and the pipeline in §2.

1. **Ground truth.** Download `https://www.federalreserve.gov/econresdata/feds/2015/files/feds2015059.zip` and parse the two HTML tables: the 44 raw indicator series (quarterly, 1990Q1–2014Q4) and the published aggregate plus 14 component indices on $(0, 1)$.

2. **Build each series** per §4, aligning every series to calendar quarters (quarterly means for daily/monthly inputs; year-on-year changes where specified). FRED series come from `https://fred.stlouisfed.org/graph/fredgraph.csv?id=SERIES_ID`; **download with `curl`, not Python `urllib`**—the FRED server stalls non-browser user agents and the download will hang. The EBP/GZ file comes from `https://www.federalreserve.gov/econres/notes/feds-notes/ebp_csv.csv` (monthly, columns `date`, `gz_spread`, `ebp`, `est_prob`, from 1973M1). Shiller's E/P from `http://www.econ.yale.edu/~shiller/data/ie_data.xls`. Y-9C items from the Federal Reserve Bank of Chicago's bulk BHCF downloads; Call Report items from

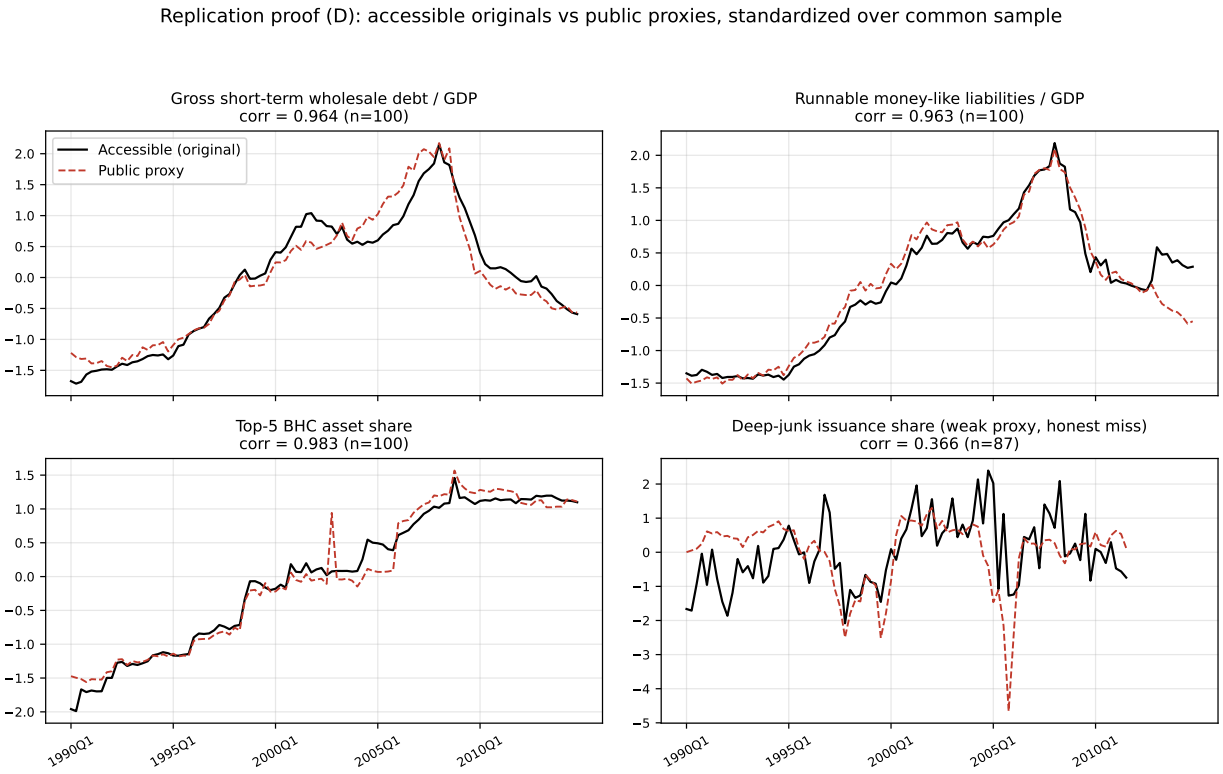


Figure 4: Replication proof (D): short-term funding, concentration, and the honest weak proxy (deep-junk issuance share, panel 4).

the FFIEC Central Data Repository bulk downloads (note: bulk files carry MDRM codes with no descriptive labels—identify items by code); SLOOS historical XML from the Federal Reserve Board's SLOOS page; origination FICO from Fannie Mae's credit supplements; CRE prices from the BIS property price statistics; NIPA items from the BEA; Z.1 mnemonics from the Federal Reserve Financial Accounts (also on FRED under `BOGZ1`-prefixed codes).

3. **Score each proxy** against the accessible raw series on the *transformed* signal (sign, transform, standardization per §2), over the overlap. Expect the correlations in §4; large deviations mean a construction error.

4. **Run the pipeline** (§2) on the 43 proxies, **masking each proxy to the accessible raw series' coverage** before transforming—without this, longer proxy histories change component composition and the comparison is invalid. Standardize each series over its full available sample; the `ma10` transform uses a 40-quarter rolling mean with minimum 20 observations. The kernel CDF uses the Silverman bandwidth on each component series; the aggregate is the mean of the 14 $(0, 1)$ components, kernel-CDF'd again.

5. **Verify** against Tables 2 and 3: aggregate 0.97 on the pipeline check, 0.93 on the full proxy panel; per-component correlations should match to the decimals quoted.

6. **Reproduce the figure:** map each component's $(0, 1)$ value to a blue-to-red color scale per quarter; the rebuilt figure should match the paper's.

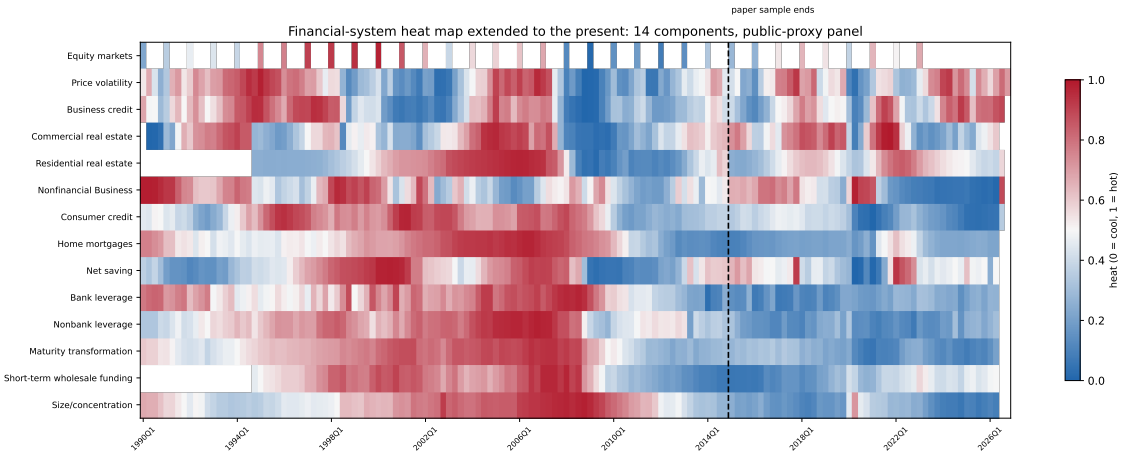


Figure 5: The financial-system heat map extended to the present, from the 43 public proxies. Blue is cool, red is hot; the dashed line marks the end of the paper's sample.

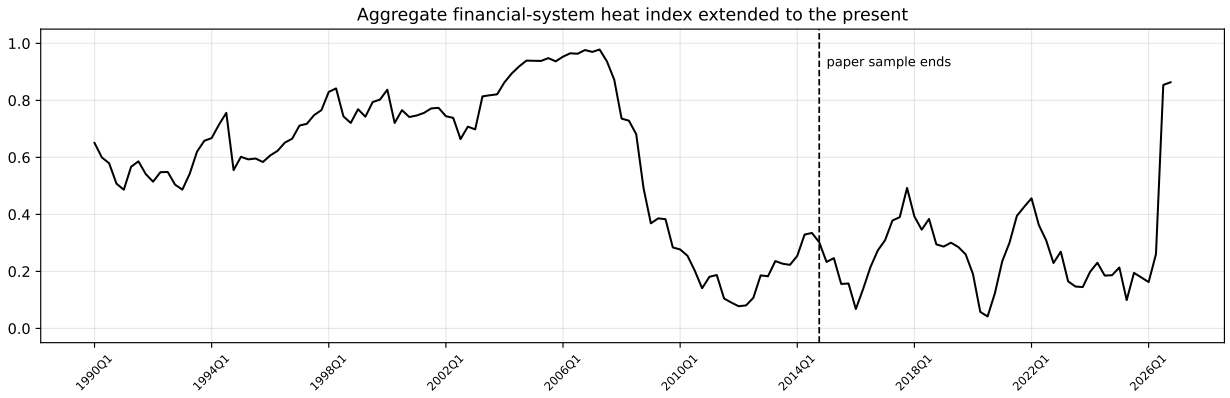


Figure 6: Aggregate financial-system heat index, 1990Q1–2026Q4, from the public-proxy panel.

Gotchas collected along the way: FRED stalls non-browser user agents (use `curl`'s default UA); the FFIEC bulk files have no descriptive labels (MDRM codes only); mask proxies to accessible coverage before the pipeline; the `ma10` transform drops the first up to 19 quarters; Basel III breaks Y-9C HC-R comparability in 2014Q1 (use the pre-2014 mnemonics and stop at 2013Q4); the SLOOS CRE aggregate must be re-dated back one quarter to match the paper's reference-quarter convention; current public vintages are revised data while the paper used 2015 vintages—expect small deviations everywhere from revision alone.

9 Limitations

- **Data vintages.** Everything above compares current-vintage public data against the paper's 2015-vintage tables. Revisions explain part of every deviation; they cannot be removed without a 2015-vintage archive (e.g. ALFRED).
- **The CRE pipeline discrepancy** (§6.3) means the CRE component cannot be fully reconciled from the paper's documentation—a finding about the paper, not the proxies.
- **The maturity gap** has no public substitute; any live monitoring implementation needs

licensed Call Report microdata or a redefined indicator.

- **Proxy dynamics $\neq$ proxy levels.** Several proxies match shape but not level (loans/deposits runs $\approx$18–20pp low; short-term money share $\approx$15% vs. $\approx$21–38%). This is by design: the pipeline standardizes, so only dynamics matter—but anyone reusing a proxy *outside* this pipeline must not mistake it for the level.
- **Coverage ends 2014Q4** by construction (the paper's sample). The public proxies extend further, but scoring stops where ground truth stops.

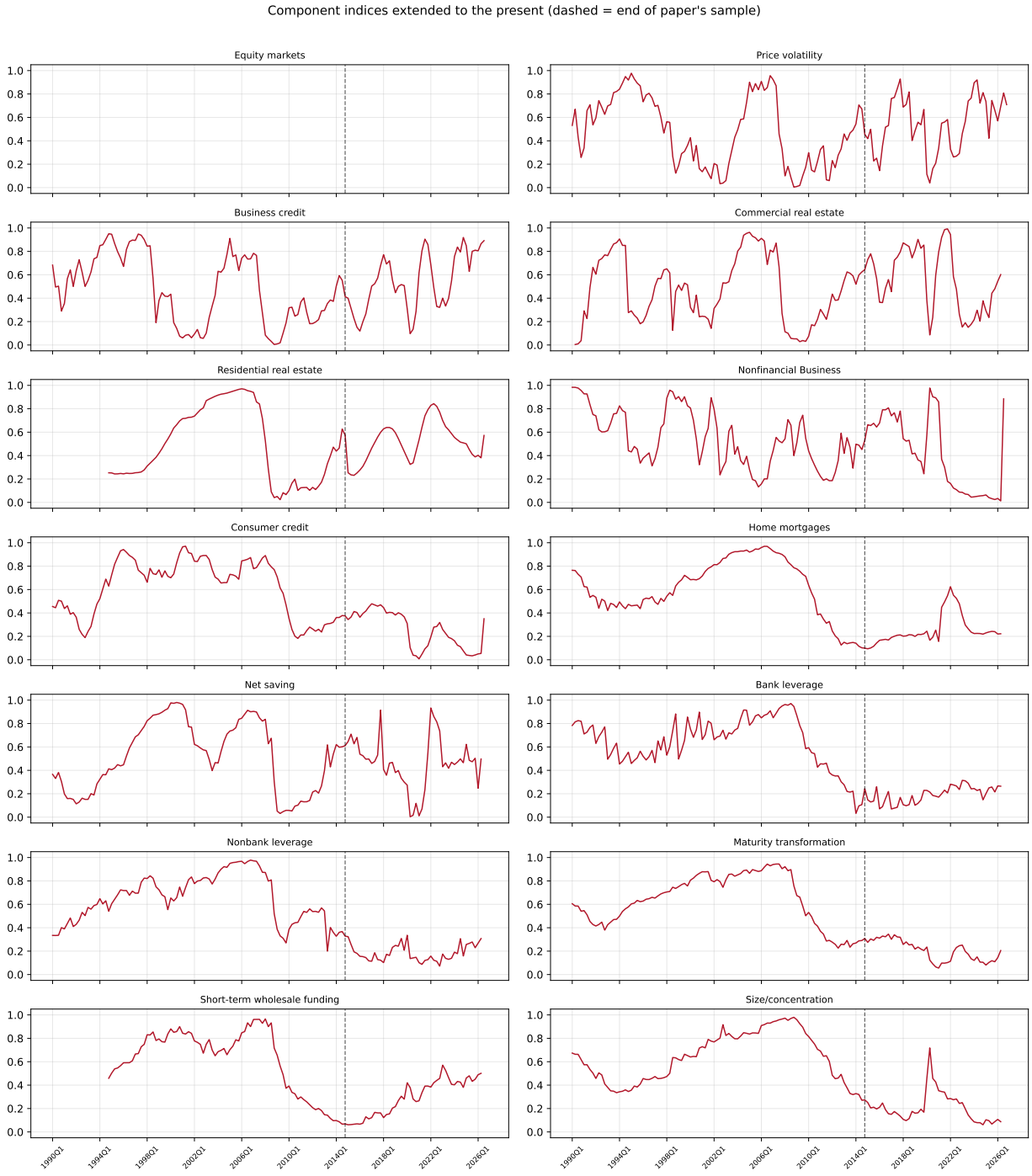


Figure 7: All 14 component indices extended to the present.